

Introduction

SCHWESERNOTES - BOOK 2

Before studying this reading, ensure you have reviewed capitalization versus expensing costs in our reading on Analyzing Income Statements and understand the implications of these approaches on the financial statements. You should also understand the various types of long-term assets, PP&E, and intangibles (from our reading on Understanding Balance Sheets and the prerequisites). Ensure you have understood the two models of recording assets in the balance sheet. Under U.S. GAAP, the cost method is required. IFRS allows companies to choose either the cost method or revaluation model for each class of asset. You should also ensure you understand depreciation and amortization methods (prerequisites).

Module 33.1: Intangible Long-Lived Assets

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LOS 33.a: Compare the financial reporting of the following types of intangible assets: purchased, internally developed, and acquired in a business combination.

Intangible assets are long-term assets that lack physical substance, such as patents, brand names, copyrights, and franchises. Some intangible assets have finite lives, while others have indefinite lives.

The cost of a finite-lived intangible asset is amortized over its useful life. Indefinite-lived intangible assets are not amortized, but they are tested for impairment at least annually. If impaired, the reduction in value is recognized in the income statement as a loss in the period in which the impairment is recognized.

Intangible assets are also considered either identifiable or unidentifiable. Under IFRS, an **identifiable intangible asset** must be the following:

- Capable of being separated from the firm or arise from a contractual or legal right
- Controlled by the firm
- Expected to provide future economic benefits

In addition, the future economic benefits must be probable, and the asset's cost must be reliably measurable.

An **unidentifiable intangible asset** is one that cannot be purchased separately and may have an indefinite life. The most common example of an unidentifiable intangible asset is goodwill. Goodwill is the excess of purchase price over the fair value of the identifiable assets (net of liabilities) acquired in a business combination.

Not all intangible assets are reported on the balance sheet. Accounting for an intangible asset depends on whether the asset was created internally, purchased externally, or obtained as part of a business combination.

Intangible Assets Created Internally

With some exceptions, costs to create intangible assets are expensed as incurred. Important exceptions are research and development costs (under IFRS) and software development costs.

Under IFRS, **research costs**, which are costs aimed at the discovery of new scientific or technical knowledge and understanding, are expensed as incurred. However, **development costs** may be capitalized. Development costs are incurred to translate research findings into a plan or design of a new product or process. To recognize an intangible asset in development, a firm must show that it can complete the asset and intends to use or sell the completed asset, among other criteria.

Under U.S. GAAP, both research and development costs are generally expensed as incurred. However, the costs of creating software for sale to others are treated in a manner similar to the treatment of research and development costs under IFRS. Costs incurred to develop software for sale to others are expensed as incurred until the product's technological feasibility has been established, after which the costs of developing a salable product are capitalized. Costs incurred developing software for internal use must be expensed until it is probable that the firm will complete the project and use the software as intended. Technological feasibility is more difficult to demonstrate than the probability of completing and using the software.

Purchased Intangible Assets

Like tangible assets, an intangible asset purchased from another party is initially recorded on the balance sheet at cost, typically its fair value at acquisition.

If the intangible asset is purchased as part of a group, the total purchase price is allocated to each asset on the basis of its fair value. For analytical purposes, an analyst is usually more interested in the type of asset acquired rather than the value assigned on the balance sheet. For example, recently acquired franchise rights may provide insight into the firm's future operating performance. In this case, the allocation of cost is not as important.

The financial statement effects of capitalizing intangible assets are the same as the effects of capitalizing other expenditures. Capitalizing results in higher net income in the first year and lower net income in the subsequent years. Similarly, assets, equity, and operating cash flow are all higher when expenditures are capitalized.

A company that generates its intangible assets internally will have lower balance sheet assets than a company that purchases its intangibles. Internally generated intangibles are typically expensed, while purchased intangibles are capitalized. This can have a considerable impact on financial statements and ratios.

Intangible Assets Obtained in a Business Combination

The **acquisition method** is used to account for business combinations. Under the acquisition method, the purchase price is allocated to the identifiable assets and liabilities of the acquired firm on the basis of fair value. This will involve establishing the fair value of any intangibles that the acquired company had internally developed and including them in the balance sheet at the date of acquisition. To capitalize these intangible assets, which had previously expensed as incurred, the asset must be an identifiable intangible asset. Any remaining amount of the purchase price is recorded as **goodwill** (see **Goodwill**). Goodwill is said to be an unidentifiable asset that cannot be separated from the business itself.

Goodwill

Purchase price	X
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Fair market value of net assets acquired	(X)
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Fair market value of identifiable intangibles previously not recognized	(X)
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Goodwill	X
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Only goodwill created in a business combination is capitalized on the balance sheet. The costs of any internally generated "goodwill" are expensed in the period incurred.

Module 33.2: Impairment and Derecognition

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LOS 33.b: Explain and evaluate how impairment and derecognition of property, plant, and equipment and intangible assets affect the financial statements and ratios.

Depreciation and amortization represent the spreading of an asset's cost to match the benefits earned over an asset's life, and they cause the balance sheet carrying value to decline. An **impairment** is an unanticipated decline in an asset's value, causing it to fall below the carrying value. **Derecognition** occurs when an asset is disposed of or retired.

Impairments

Both IFRS and U.S. GAAP require firms to write down impaired assets by recognizing a loss in the income statement, but the standards are applied slightly differently. If the impairment loss is material in size and nature, it will be shown as an unusual or infrequent item.

Professor's Note

The following discussion applies to both tangible and intangible long-lived assets with finite lives that are held for use.

Impairments Under IFRS

Under IFRS, the firm must annually assess whether events or circumstances indicate that an impairment of an asset's value has occurred. For example, there may have been a significant decline in the market value of the asset, or a significant change in the asset's physical condition. If so, the asset's value must be tested for impairment.

An asset is impaired when its carrying value (original cost less accumulated depreciation) exceeds the **recoverable amount**. The recoverable amount is the greater of its fair value less any selling costs and its **value in use**. The value in use is the present value of its future cash flow stream from continued use and disposal. Value in use is a highly subjective figure requiring estimation of future cash flows, disposal proceeds, and the selection of an appropriate discount rate.

If impaired, the asset's value must be written down on the balance sheet to the recoverable amount. An impairment loss, equal to the excess of carrying value over the recoverable amount, is recognized in the income statement.

Under IFRS, an impairment loss on an identifiable long-lived asset can be reversed if the asset's value recovers in the future. However, the loss reversal is limited to the original impairment loss.

Impairments Under U.S. GAAP

Under U.S. GAAP, an asset is tested for impairment only when events and circumstances indicate the firm may not be able to recover the carrying value through future use.

Determining an impairment and calculating the loss potentially involve two steps. In the first step, the asset is tested for impairment by applying a **recoverability test**. If the asset is impaired, the second step involves measuring the loss.

Regarding *recoverability*, an asset is considered impaired if the carrying value (original cost less accumulated depreciation) is greater than the asset's future *undiscounted* cash flow stream. Because the recoverability test is based on estimates of future undiscounted cash flows, tests for impairment involve considerable management discretion. Like value in use under IFRS, estimating the future undiscounted cash flows from an asset's use and disposal is highly subjective.

Regarding *loss measurement*, if impaired, the asset's value is written down to fair value on the balance sheet—and a loss, equal to the excess of carrying value over the fair value of the asset (or the *discounted* value of its future cash flows, if the fair value is not known), is recognized in the income statement. The discounted value of future cash flows is equivalent of value in use under IFRS.

Under U.S. GAAP, loss recoveries are typically not permitted.

Professor's Note

The difference between testing for impairment and measuring the impairment loss can be confusing. In testing for impairment under U.S. GAAP, undiscounted cash flows are used. Once impairment has been detected, the loss is based on fair value, or the discounted expected future cash flows. Using undiscounted cash flows to test for impairment keeps PP&E assets from becoming "impaired" by increases in the discount rate when interest rates increase. In measuring the impairment loss, U.S. GAAP uses fair value, not fair value less selling costs.

EXAMPLE: Asset impairment

Information related to equipment owned by Brownfield Company follows:

Original cost	\$900,000
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Accumulated depreciation to date	\$100,000
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Expected future cash flows	\$795,000
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Fair value	\$790,000
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Value in use	\$785,000
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Selling costs	\$30,000
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Assuming Brownfield will continue to use the equipment, test the asset for impairment under both IFRS and U.S. GAAP and discuss the results.

Answer:

The carrying value of the equipment is \$900,000 original cost – \$100,000 accumulated depreciation = \$800,000, and the recoverable amount under IFRS is \$785,000 (greater of \$785,000 value in use and \$760,000 fair value less selling costs). Under IFRS, the asset is written down on the balance sheet to the \$785,000 recoverable amount, and a \$15,000 loss (\$800,000 carrying value – \$785,000 recoverable amount) is recognized in the income statement.

Under U.S. GAAP, the asset is impaired because the \$795,000 expected future cash flows is less than the \$800,000 carrying value. The asset is written down on the balance sheet to its \$790,000 fair value, and a \$10,000 loss (\$800,000 carrying value – \$790,000 fair value) is recognized on the income statement.

Impairment reduces an asset's carrying value on the balance sheet. An impairment charge is recognized as a loss in the income statement, reducing assets and equity (retained earnings). In the year of impairment, return on assets (ROA) and return on equity (ROE) will decrease because the impairment charge reduces net income.

In subsequent periods, net income will be higher than it would have been without the impairment charge because depreciation or amortization will be lower (the asset has a lower depreciable value). Both ROA and ROE will increase in periods after the impairment charge because both equity and assets will fall as a result of the impairment charge. Asset turnover will increase in the period in which the impairment charge is taken, and in subsequent periods.

Asset impairment has no impact on cash flow because the impairment does not reduce taxable income; it is an unrealized loss until the asset is disposed of.

Analysis of Impairments

An impairment loss essentially indicates that the firm has not recognized sufficient depreciation or amortization expense, and has overstated earnings as a result.

The judgment required in determining asset impairments gives management considerable discretion about the timing and amounts of impairment charges. Consequently, impairment decisions present an opportunity for management to manipulate earnings. Waiting to recognize an impairment loss until a period of relatively high earnings would tend to smooth earnings.

Alternatively, existing managements may take more impairment charges in periods when earnings will be poor due to external (macroeconomic or industry) factors. New managements may also choose to take more or greater impairment

charges when they take over. In either case, the resulting low earnings might not be perceived as the "fault" of management, and lower values for assets and equity give a boost to ROA and ROE going forward.

Intangible Assets With Indefinite Lives

Intangible assets with indefinite lives are not amortized; rather, they are tested for impairment at least annually. An impairment loss is recognized when the carrying amount exceeds fair value.

Long-Lived Assets Held for Sale

If a firm intends to sell an asset, it is probable that the asset will be sold, and the asset is available for immediate sale, then it must be reclassified from *held for use* to *held for sale*. When a firm reclassifies an asset as held for sale, the asset is no longer depreciated or amortized. The held-for-sale asset is impaired if its carrying value exceeds its fair value less selling costs. If impaired, the asset is written down to net realizable value, and the loss is recognized in the income statement.

For long-lived assets held for sale, the loss can be reversed under IFRS and U.S. GAAP if the value of the asset recovers in the future. However, the loss reversal is limited to the original impairment loss. Thus, the carrying value of the asset after reversal cannot exceed the carrying value before the impairment was recognized.

Derecognition

Eventually, long-term assets are removed from the balance sheet. Derecognition occurs when assets are sold, exchanged, or abandoned.

When a long-term asset is sold, the asset is removed from the balance sheet and the difference between the sale proceeds and the carrying value of the asset is reported as a gain or loss in the income statement, as seen in

Derecognition of Long-Term Assets. The carrying value is equal to original cost minus accumulated depreciation and any impairment charges.

Derecognition of Long-Term Assets

Disposal proceeds	X	Cash flow from investing
Carrying value before disposal	(X)	Removed from balance sheet
Disposal gain (loss)	X/(X)	Accounting gain or (loss) in income statement

The gain or loss is reported in the income statement with other gains and losses, or reported separately as an unusual or infrequent item if material. Also, if the firm presents its cash flow statement using the indirect method, the gain or loss is removed from net income to compute cash flow from operations because the proceeds from selling a long-lived asset are an investing cash inflow.

If a long-lived asset is abandoned, the treatment is similar to a sale, except there are no proceeds. In this case, the carrying value of the asset is removed from the balance sheet, and a loss of that amount is recognized in the income statement.

If a long-lived asset is exchanged for another asset, a gain or loss is computed by comparing the carrying value of the old asset with fair value of the old asset (or the fair value of the new asset, if that value is clearly more evident). The carrying value of the old asset is removed from the balance sheet, and the new asset is recorded at its fair value. If no reliable measure of fair value exists, the new asset is valued at the carrying value of the old asset.

A **spinoff** is the transfer of assets that constitute an entire division or subsidiary into a new legal entity, upon which the shares of the spinnee are subsequently distributed to its shareholders, and the shareholders do not surrender any stock of the spinnor.¹ See **Spinoff** as follows.

Spinoff

Before Spinoff

Shareholders' Company A



Company A



Company B

After Spinoff

Shareholders' Company A



Company A



Company B

Before the spinoff, Company A (the spinnor) owns Company B (the spinnee), and shareholders just have shares in Company A. After the spinoff, the shareholders have shares in both Company A and Company B. Company B is no longer included in the consolidated accounts of Company A.

Once the spinoff becomes probable, the balance sheet assets and liabilities of the spinnee will be transferred from assets held for use to assets held for sale in the spinnor's accounts. Assets held for sale may be referred to as assets held for distribution, in this context. No profit or loss is recorded on the disposal in the income statement.

¹Definition per U.S. GAAP, ASC 505-60.

Module 33.3: Long-Term Asset Disclosures

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LOS 33.c: Analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets.

IFRS Disclosures

Under IFRS, the firm must disclose the following for each class of property, plant, and equipment (PP&E):

- Basis for measurement (usually historical cost)
- Depreciation method
- Depreciation expense in the period
- Useful lives or depreciation rate
- Gross carrying value and accumulated depreciation at the start and end of the period
- Reconciliation of carrying amounts from the beginning of the period to the end of the period

The firm must also disclose the following:

- Title restrictions and assets pledged as collateral
- Agreements to acquire PP&E in the future

If the revaluation (fair value) model is used, the firm must disclose the following:

- The revaluation date
- How fair value was determined

- Carrying value using the historical cost model
- Revaluation surplus in other comprehensive income (OCI)

Under IFRS, the disclosure requirements for intangible assets are similar to those for PP&E, except that the firm must disclose whether the useful lives are finite or indefinite.

For impaired assets, the firm must disclose the following:

- Amounts of impairment losses and reversals by asset class
- Where the losses and loss reversals are recognized in the income statement
- Circumstances that caused the impairment loss or reversal

U.S. GAAP Disclosures

Under U.S. GAAP, the PP&E disclosures include the following:

- Depreciation expense by period
- Balances of major classes of assets by nature and function, such as land, improvements, buildings, machinery, and furniture
- Accumulated depreciation by major classes, or in total
- General description of depreciation methods used

Under U.S. GAAP, the disclosure requirements for intangible assets are similar to those for PP&E. In addition, the firm must provide an estimate of amortization expense for the next five years.

For impaired assets, the firm must disclose the following:

- A description of the impaired asset
- Circumstances that caused the impairment
- How fair value was determined
- The amount of loss
- Where the loss is recognized in the income statement

Under IFRS, the depreciation and amortization expenses may be shown on the face of the income statement if producing an income statement using the nature of the expense approach. If the company prepares the income statement using the function of the expense method, it will not be shown on the face; instead, it will be included in the cost of sales and sales general and admin. If the indirect method is used to produce cash flow from operations, depreciation and amortization will appear as noncash charges. For companies adopting the direct method, depreciation will not appear in the computation of operating cash flows; however, U.S. GAAP requires the indirect method reconciliation of net income to operating cash flows to be disclosed in the footnotes. Cash received on asset disposals and spent on asset acquisitions are presented as cash flows from investing activities.

Analyzing Long-Lived Asset Disclosures

Fixed-asset turnover shows revenue per dollar of fixed assets and indicates how efficiently a firm is using its long-term assets to generate sales. A higher ratio is interpreted as greater efficiency.

$$\text{fixed asset turnover} = \text{revenue} / \text{average fixed assets}$$

Financial statement disclosures provide an analyst considerable information about a company's fixed assets and depreciation (amortization) methods. An analyst can use the data to estimate the average age of the firm's assets. The average age is useful for two reasons:

1. Older, less-efficient assets may make a firm less competitive.
2. The average age of assets helps an analyst to estimate the timing of major capital expenditures and a firm's future financing requirements.

The level of detail provided in footnote disclosures regarding fixed assets and depreciation varies across firms. Because assets are often grouped by their useful lives, the following methods of estimating the average age, economic life, and remaining useful life of a firm's assets do not produce precise values, but they can highlight issues for further investigation. The following computations assume zero salvage values and straight-line depreciation, and as a result, they are less insightful for asset classes subject to accelerated depreciation methods and substantial salvage values.

Three useful calculations (in years) for an analyst are as follows.

Average Age

$$\text{average age} = \frac{\text{accumulated depreciation}}{\text{annual depreciation expense}}$$

This calculation is more accurate for a firm that uses straight-line depreciation. The calculation can be significantly affected by the mix of assets.

Total Useful Life

$$\text{total useful life} = \frac{\text{historical cost (gross cost)}}{\text{annual depreciation expense}}$$

Historical cost is gross PP&E before deducting accumulated depreciation.

Remaining Useful Life

$$\text{remaining useful life} = \frac{\text{ending net PP\&E}}{\text{annual depreciation expense}}$$

Net PP&E is equal to original cost (gross PP&E) minus accumulated depreciation.

Professor's Note

The remaining useful life can also be approximated by subtracting the average age from the total useful life.

EXAMPLE: Calculating average age and total useful life

At the end of 20X8, a company has gross PP&E of \$3 million and accumulated depreciation of \$1 million. During the year, the depreciation expense was \$500,000.

What is the average age, total useful life, and remaining useful life of the company's PP&E?

Answer:

$$\text{average age} = \frac{\text{accumulated depreciation}}{\text{depreciation expense}} = \frac{\$1,000,000}{\$500,000} = 2 \text{ years}$$

$$\text{total useful life} = \frac{\text{historical cost}}{\text{depreciation expense}} = \frac{\$3,000,000}{\$500,000} = 6 \text{ years}$$

$$\text{remaining useful life} = \frac{\text{ending net PP\&E}}{\text{depreciation expense}} = \frac{\$2,000,000}{\$500,000} = 4 \text{ years}$$

Another popular metric is the ratio of annual capital expenditures to depreciation expense. This ratio provides information about whether the firm is maintaining its production capacity by replacing its PP&E at the same rate as its assets are being depreciated.

Reading 33: Key Concepts

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LOS 33.a

The cost of a purchased finite-lived intangible asset is amortized over its useful life. Indefinite-lived intangible assets are not amortized, but they are tested for impairment at least annually. The cost of internally developed intangible assets is expensed.

Under IFRS, research costs are expensed, but development costs may be capitalized. Under U.S. GAAP, both research and development costs are expensed as incurred, except in the case of software.

The acquisition method is used to account for assets acquired in a business combination. The purchase price is allocated to the fair value of identifiable assets of the acquired firm less its liabilities. Any excess of the purchase price above the fair value of the acquired firm's net assets is recorded as goodwill, an unidentifiable intangible asset that cannot be separated from the business itself.

Compared to expensing an asset's cost, capitalization results in the following:

- Lower expense and higher net income in period of acquisition, and higher expense (depreciation or amortization) and lower net income in each of the remaining years of the asset's life
- Higher assets and equity
- Lower CFI and higher CFO because the cost of a capitalized asset is classified as an investing cash outflow
- Higher ROE and ROA in the initial period, and lower ROE and ROA in subsequent periods because net income is lower, and both assets and equity are higher
- Lower debt-to-assets and debt-to-equity ratios because assets and equity are higher

LOS 33.b

Impairment charges decrease net income, assets, and equity, which results in lower ROA and ROE and higher debt-to-equity and debt-to-assets ratios for a typical firm.

Under IFRS, an asset is impaired when its carrying value exceeds the recoverable amount. The recoverable amount is the greater of fair value less selling costs and the value in use (present value of expected cash flows). If impaired, the asset is written down to the recoverable amount. Loss recoveries are permitted, but not above historical cost.

Under U.S. GAAP, an asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows. If impaired, the asset is written down to its fair value. Subsequent recoveries are not allowed for assets held for use.

Asset impairments result in losses in the income statement. Impairments have no impact on cash flow, as they have no tax or other cash flow effects until disposal of the asset.

Derecognition of assets can result in either a gain or loss on the income statement. A loss will reduce net income and assets, while a gain will increase net income and assets.

When a long-lived asset is *sold*, the difference between the sale proceeds and the carrying (book) value of the asset is reported as a gain or loss in the income statement.

When a long-lived asset is *abandoned*, the carrying value is removed from the balance sheet, and a loss is recognized in that amount.

If a long-lived asset is *exchanged* for another asset, a gain or loss is computed by comparing the carrying value of the old asset with fair value of the old asset (or fair value of the new asset, if more clearly evident).

LOS 33.c

There are differences in the disclosure requirements for long-lived assets under IFRS and U.S. GAAP. However, firms are generally required to disclose the following:

- Carrying values for each class of asset
- Accumulated depreciation and amortization
- Title restrictions and assets pledged as collateral

- For impaired assets, the loss amount and the circumstances that caused the loss
- For revalued assets (IFRS only), the revaluation date, how fair value was determined, and the carrying value using the historical cost model

Analysts can use disclosures of the historical cost, accumulated depreciation (amortization), and annual depreciation (amortization) expense to estimate average age of assets, total useful life of assets, and remaining useful life of assets.

$$\begin{aligned}
 \text{average age} &= \frac{\text{accumulated depreciation}}{\text{annual depreciation expense}} \\
 \text{total useful life} &= \frac{\text{historical cost}}{\text{annual depreciation expense}} \\
 \text{remaining useful life} &= \frac{\text{ending net PP\&E}}{\text{annual depreciation expense}}
 \end{aligned}$$